Preview: y2007_q23 (v1)

Counting digits by groups (1-, 2-, 3-digit numbers); how many numbers are in a range

Part A: Each question is worth 3 points

1. How many digits are needed to write down all the numbers from 1 to 100?

A. 100 ***B.*** 192 ***C.*** 193 ***D.*** 200 ***E.*** 300

2. How many digits are needed to write down all the numbers from 1 to 300?

A. 300 ***B.*** 600 ***C.*** 789 ***D.*** 790 ***E.*** 792

3. Mia writes all the numbers from 90 to 101.
How many digits does Mia write?

A. 22 ***B.*** 24 ***C.*** 25 ***D.*** 26 ***E.*** 36

4. Leo writes all the numbers from 992 to 1010.
How many digits does Leo write?

A. 68 ***B.*** 69 ***C.*** 72 ***D.*** 76 ***E.*** 82

Part B: Each question is worth 4 points

5. All the numbers from 1 to 100 are written down.
How many times is the digit 0 written?

A. 7 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 11

6. All the numbers from 1 to 300 are written down.
How many times is the digit 0 written?

A. 51 ***B.*** 52 ***C.*** 59 ***D.*** 60 ***E.*** 62

7. Only the even numbers from 2 to 100 are written down: 2, 4, 6, ..., 100.
How many digits are written?

A. 50 ***B.*** 95 ***C.*** 96 ***D.*** 97 ***E.*** 100

8. Only the even numbers from 2 to 200 are written down: 2, 4, 6, ..., 200.
How many digits are written?

A. 200 ***B.*** 247 ***C.*** 249 ***D.*** 262 ***E.*** 280

Part C: Each question is worth 5 points

9. Leo types the numbers from 1 to 50 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,50.
How many keys does Leo press (each digit and each comma is one key)?

A. 141 ***B.*** 140 ***C.*** 139 ***D.*** 100 ***E.*** 91

10. Mia types the numbers from 1 to 150 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,150.
How many keys does Mia press (each digit and each comma is one key)?

A. 300 ***B.*** 342 ***C.*** 490 ***D.*** 491 ***E.*** 500

Answer Key

1. **B**	2. **E**	3. **D**	4. **A**	5. **E**	6. **A**
7. **D**	8. **B**	9. **B**	10. **D**		

Solutions

1. B (192) — Numbers with 1, 2, 3, ... digits: 9 × 1 + 90 × 2 + 1 × 3 = 192. (There are 90 two-digit numbers: 10 to 99.) [2007 Q23 · one_to_round]

2. E (792) — Numbers with 1, 2, 3, ... digits: 9 × 1 + 90 × 2 + 201 × 3 = 792. (There are 90 two-digit numbers: 10 to 99.) [2007 Q23 · one_to_round]

3. D (26) — There are 12 numbers: 10 × 2 + 2 × 3 = 26 digits. [2007 Q23 · across_hundred]

4. A (68) — There are 19 numbers: 8 × 3 + 11 × 4 = 68 digits. [2007 Q23 · across_hundred]

5. E (11) — A 0 at the end: 10, 20, ..., 100: 10 numbers. Plus the zeros in the middle of 100-100 (like 101 or 100 itself): 1 more. Total 11. [2007 Q23 · count_zeros]

6. A (51) — A 0 at the end: 10, 20, ..., 300: 30 numbers. Plus the zeros in the middle of 100-300 (like 101 or 100 itself): 21 more. Total 51. [2007 Q23 · count_zeros]

7. D (97) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 1. 4 × 1 + 45 × 2 + 1 × 3 = 97. [2007 Q23 · even_only]

8. B (247) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 51. 4 × 1 + 45 × 2 + 51 × 3 = 247. [2007 Q23 · even_only]

9. B (140) — Digits: 9 × 1 + 41 × 2 = 91. Commas: one fewer than the numbers, 49. 91 + 49 = 140. [2007 Q23 · with_commas]

10. D (491) — Digits: 9 × 1 + 90 × 2 + 51 × 3 = 342. Commas: one fewer than the numbers, 149. 342 + 149 = 491. [2007 Q23 · with_commas]